Cryptography and Network Security Midterm Exam Question Bank B.E. (AI&DS)

Unit-1

1. **Explain the various types of security attacks in a computer network.** Discuss security services and security mechanisms used to protect information and communication systems.
2. **Explain the fundamentals of cryptography and the conventional encryption model.** Describe the components of the model and explain the role of plaintext, Ciphertext, encryption algorithm, decryption algorithm, and key.
3. **Explain classical encryption techniques in detail.** Describe **substitution ciphers and transposition ciphers** with suitable examples. Compare their working principles.
4. **What is cryptanalysis?** Explain the objectives and general techniques of cryptanalysis. Also, discuss **steganography** and differentiate between steganography and cryptography.
5. **Differentiate between stream ciphers and block ciphers.** Explain their working principles, characteristics, advantages, limitations, and applications with suitable examples.
6. **Explain the principles of modern block ciphers.** Discuss **Shannon's theory of confusion and diffusion** and explain the significance of the **Feistel structure** in the design of block ciphers.
7. **Explain the Data Encryption Standard (DES) in detail.** Describe its overall structure, key generation process, Feistel rounds, S-boxes, and encryption/decryption process with a neat diagram.
8. **Discuss the strength and security of DES.** Explain the concepts of **differential cryptanalysis and linear cryptanalysis** and analyze their impact on the security of DES.
9. **Explain the different modes of operation of block ciphers.** Discuss **ECB, CBC, CFB, OFB, and CTR modes**, highlighting their working, advantages, limitations, and suitable applications.
10. **Explain Triple DES (3DES) and the Advanced Encryption Standard (AES).** Describe their structures, key sizes, block sizes, number of rounds, and security features. **Compare DES, 3DES, and AES** with respect to security and performance.

Unit-2

1. **Explain the concept of confidentiality using conventional encryption.** Discuss the conventional encryption model and explain how traffic confidentiality can be achieved in a communication network.
2. **Explain the key distribution problem in conventional encryption systems.** Describe different methods of key distribution and discuss their advantages and limitations.
3. **What is random number generation?** Explain the requirements of a good random number generator and discuss its importance in cryptographic applications.
4. **Define a group, ring, and field.** Explain their fundamental properties with suitable examples, and differentiate between a ring and a field.
5. **Explain prime numbers and relatively prime numbers.** How are they used in cryptographic algorithms? Illustrate with suitable examples.
6. **Explain modular arithmetic in detail.** Discuss modular addition, subtraction, multiplication, and division with suitable numerical examples.
7. **State and prove Fermat's Little Theorem and Euler's Theorem.** Explain how these theorems are applied in cryptographic algorithms.
8. **Explain primality testing techniques used in cryptography.** Discuss the importance of testing whether a large integer is prime and describe any suitable primality-testing method.
9. **Explain Euclid's Algorithm and the Extended Euclidean Algorithm.** Using a suitable example, find the greatest common divisor (GCD) of two integers and determine their modular multiplicative inverse.
10. **Explain the Chinese Remainder Theorem (CRT) and discrete algorithms.** Solve a suitable system of simultaneous congruence's using CRT and discuss its applications in cryptography.